

Partial Fraction Expansion Complex Roots

MATLAB's residue function can aid partial fraction expansion when transforming from the s-domain to the time domain. However, interpreting the results of the residue command can be complicated when the poles (roots of the denominator) are complex.

Imaginary Poles (No Real Part)

Consider a specific example where the task is to do find the inverse Laplace transform of the s-domain functions

$$F(s) = \frac{1}{(s+1)(s^2+3^2)}.$$

The denominator has three roots – 0, +3i, and -3i. We can use MATLAB to do the partial fraction expansion, but the results are challenging to interpret:

```
>> num = [1];
den = conv([1 1],[1 0 9]);
[K,p,k]=residue(num,den)
K =
    -0.0500 - 0.0167i
    -0.0500 + 0.0167i
     0.1000 + 0.0000i
p =
     0.0000 + 3.0000i
     0.0000 - 3.0000i
    -1.0000 + 0.0000i
k =
     1
```

Using this directly, we can express $F(s)$ as

$$F(s) = \frac{-0.05 - 0.0167i}{s - 3i} + \frac{-0.05 + 0.0167i}{s + 3i} + \frac{0.1}{s + 1}$$

if we do the algebra to combine the first two terms, we get

$$\begin{aligned} F(s) &= \frac{2(-0.05)s + 2(0.0167)(3)}{s^2 + 3^2} + \frac{0.1}{s + 1} \\ &= \frac{-0.1s}{s^2 + 3^2} + \frac{0.1}{s^2 + 3^2} + \frac{0.1}{s + 1}. \end{aligned}$$

So, in general, when we have imaginary roots (no real part), we can interpret the output of the residue command as follows. Given this general result

$$\begin{aligned}
 &[K, p, k] = \text{residue}(\text{num}, \text{den}) \\
 &K = \\
 &\quad \alpha - \beta i \\
 &\quad \alpha + \beta i \\
 &\quad \dots \\
 &p = \\
 &\quad +\omega i \\
 &\quad -\omega i \\
 &\quad \dots \\
 &k = \\
 &\quad []
 \end{aligned}$$

we would have a term in the resulting expression for $F(s)$

$$F(s) = \frac{2(\alpha)s + 2(\beta)(\omega)}{s^2 + \omega^2} + \dots$$

Complex Poles (Real and Imaginary Parts)

To be more general, we should consider the case where the roots of the denominator (AKA the poles) are complex, with both real and imaginary parts. In general, MATLAB may produce output with the format...

$$\begin{aligned}
 &[K, p, k] = \text{residue}(\text{num}, \text{den}) \\
 &K = \\
 &\quad \alpha - \beta i \\
 &\quad \alpha + \beta i \\
 &\quad \dots \\
 &p = \\
 &\quad -a + \omega i \\
 &\quad -a - \omega i \\
 &\quad \dots \\
 &k = \\
 &\quad []
 \end{aligned}$$

If we use this result directly we get a messy expression

$$F(s) = \frac{\alpha - \beta i}{s + (a - bi)} + \frac{\alpha + \beta i}{s + (a + bi)} + \dots$$

which we can simplify to

$$F(s) = 2 \frac{(\alpha(s+a) + \beta\omega)}{(s+a)^2 + \omega^2} + \dots$$

We can then take the inverse Laplace transform of this term...

$$f(t) = 2[\alpha e^{-at} \cos(\omega t) + \beta e^{-at} \sin(\omega t)]$$